

Stage 1: Inverse-Designed Diffractive Optical Readout for Spectrometer-Free Biosensing

Hemal Sharma

Numerical project, Stage 1 submission

Abstract

Stage 1 builds and validates the forward model of a spectrometer-free diffractive biosensor: the calculation that maps an analyte concentration c to the signals of two photodetectors. A Lorentzian resonance, shifted by the analyte, filters a broadband source; a phase-only diffractive optical element (DOE) shapes the light; angular-spectrum propagation carries every wavelength to the detector plane; and the detectors integrate the incoherent sum over wavelengths. With a linear phase grating as the DOE, the model reproduces analytic Gaussian propagation to 2×10^{-6} and the grating equation to 1.8 nm, conserves power to machine precision, and has converged numerically.

1 Introduction

Resonant biosensors report the analyte concentration as a shift of a resonance wavelength,

$$\lambda_r(c) = \lambda_0 + \alpha c, \quad (1)$$

which is normally read out with a spectrometer. This project instead places a passive diffractive element between the sensor and a few photodetectors, so that the spectral shift becomes a change of detector signal directly (Fig. 1). The DOE does not make the sensor more sensitive; it makes the small spectral change easier to detect.

Every later stage optimizes the DOE through one forward calculation, $c \rightarrow S(\lambda; c) \rightarrow \text{DOE} \rightarrow \text{propagation} \rightarrow I(x, y; c) \rightarrow I_A, I_B, R$. Stage 1 builds and validates that calculation. No optimization is performed; the DOE is a linear phase grating, whose behaviour is known exactly.

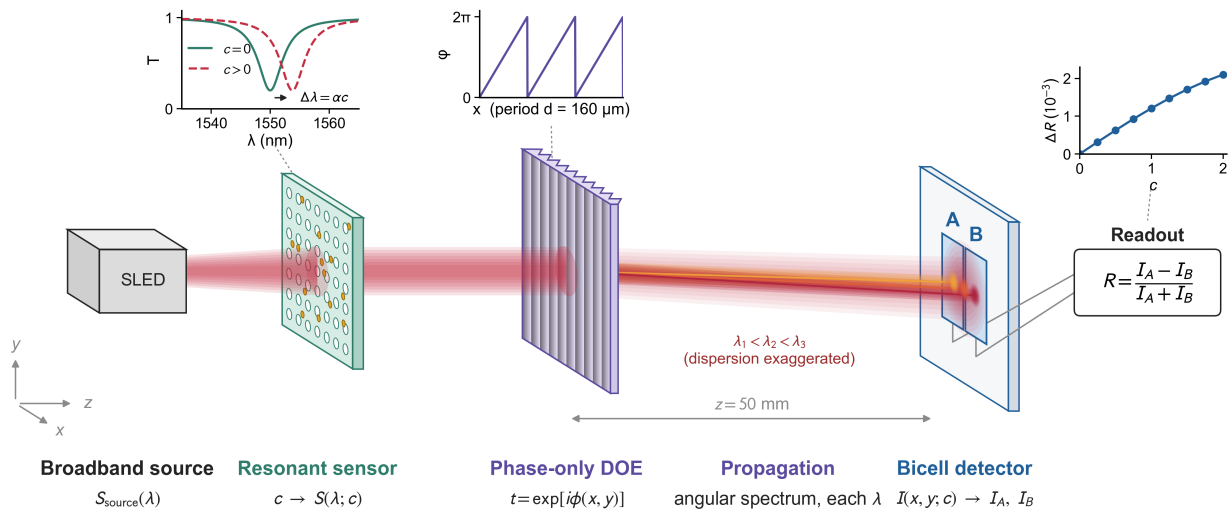


Figure 1: System schematic. A broadband source passes through the resonant sensor, whose transmission dip shifts with the analyte concentration. The phase-only DOE diffracts the light, which propagates 50 mm to two detectors A and B; their normalized difference R is the readout. No spectrometer is used.

2 Sensor model

The sensor is a Lorentzian transmission dip of depth A and half-width γ , centred at the resonance wavelength of Eq. (1), which filters the source spectrum:

$$T(\lambda; c) = 1 - \frac{A}{1 + \left(\frac{\lambda - \lambda_r(c)}{\gamma} \right)^2}, \quad (2)$$

$$S(\lambda; c) = S_{\text{source}}(\lambda) T(\lambda; c). \quad (3)$$

The resonance lies at $\lambda_0 = 1550 \text{ nm}$ with a 5 nm linewidth (FWHM), and one unit of concentration shifts it by one half-width, $\alpha = 2.5 \text{ nm}$. The source is a Gaussian of 20 nm FWHM. Figure 2 shows the dip moving with concentration and the resulting sensor output.

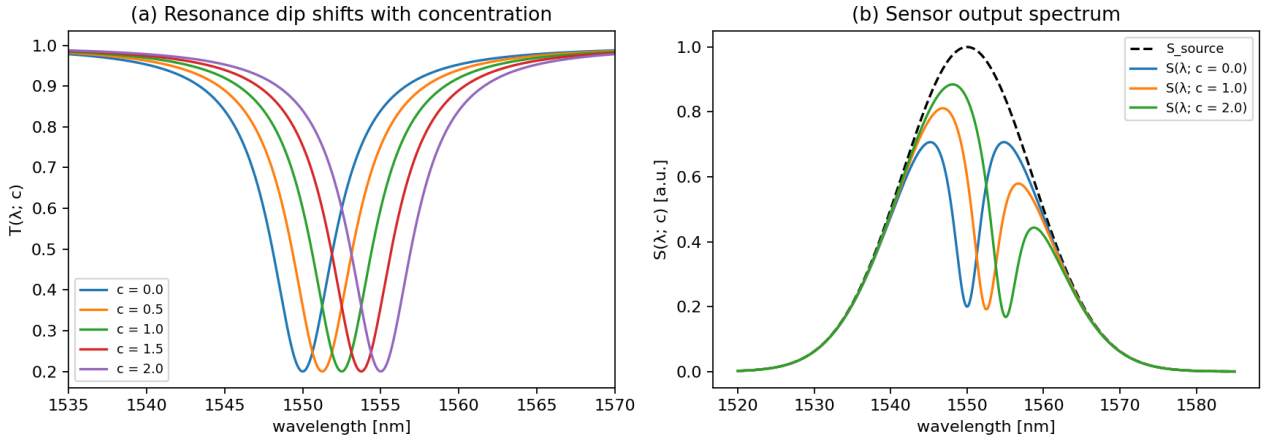


Figure 2: Sensor model. (a) The resonance dip moves to longer wavelength as the concentration increases. (b) Output spectrum: the dip cut into the Gaussian source.

3 DOE and input field

The DOE is a thin, lossless, phase-only mask of 256×256 pixels with a $10 \mu\text{m}$ pitch (2.56 mm aperture). For Stage 1 it is a linear phase grating of period $d = 160 \mu\text{m}$ (Fig. 3),

$$t(x, y) = e^{i\phi(x, y)}, \quad \phi(x, y) = \frac{2\pi x}{d} \bmod 2\pi, \quad (4)$$

which sends the light into the $+1$ diffraction order at

$$\sin \theta_1 = \lambda/d. \quad (5)$$

The DOE is illuminated by a collimated Gaussian beam of radius $w_0 = 200 \mu\text{m}$. It changes only the phase of the beam, not its amplitude or power (Fig. 4):

$$E_{\text{in}}(x, y) = E_0 e^{-(x^2+y^2)/w_0^2}, \quad E_0^+(x, y) = E_{\text{in}}(x, y) t(x, y). \quad (6)$$

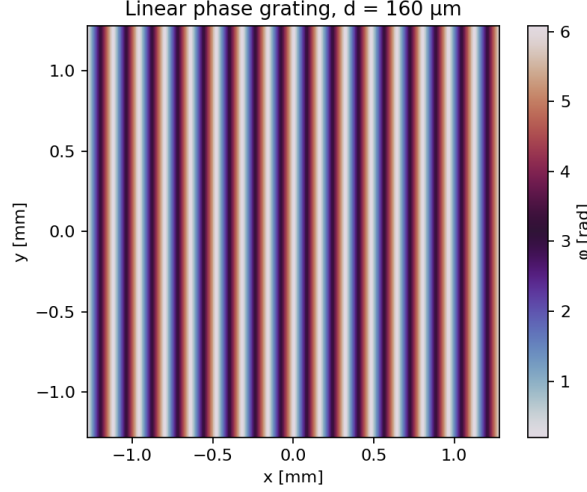


Figure 3: DOE phase map: linear phase grating over the full 2.56 mm aperture.

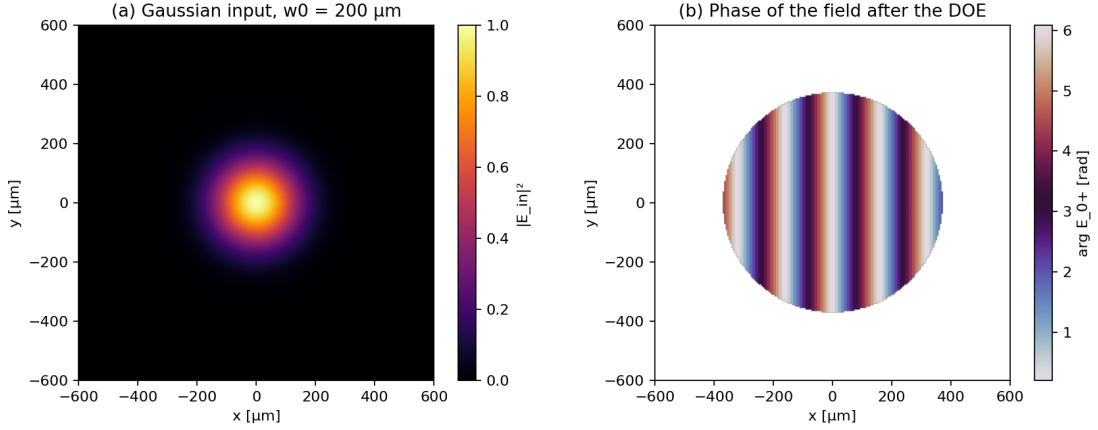


Figure 4: (a) Gaussian input intensity. (b) Phase of the field just after the DOE.

4 Propagation and detection

The field is propagated to the detector plane, $z = 50$ mm away, with the angular-spectrum method, separately at every wavelength:

$$\tilde{E}(k_x, k_y, z; \lambda) = \tilde{E}(k_x, k_y, 0) e^{ik_z z}, \quad k_z = \sqrt{k^2 - k_x^2 - k_y^2}, \quad k = \frac{2\pi}{\lambda}. \quad (7)$$

The spatial grid is the same for every wavelength, so λ enters only through k : the grating sends each wavelength to a slightly different position, which is how spectral information becomes spatial information. The method is FFT-based, exact for finite distances, and differentiable, so the same code can supply gradients for inverse design. A band-limited form [?] is used to avoid artefacts from the periodic FFT window. A Fraunhofer (far-field) model was also implemented as a cross-check; it is not accurate at 50 mm, which is shorter than the Rayleigh range, so all results use the angular spectrum.

The source wavelengths are mutually incoherent, so the detector adds intensities. With 150 wavelengths from 1520 to 1585 nm,

$$I(x, y; c) = \sum_k S(\lambda_k; c) |E(x, y; \lambda_k)|^2 \Delta\lambda. \quad (8)$$

Two detectors, A and B ($300 \times 600 \mu\text{m}$ each), sit either side of the diffracted spot. Their normalized difference R is insensitive to changes in source power:

$$I_{A,B}(c) = \iint_{A,B} I(x, y; c) dx dy, \quad R(c) = \frac{I_A - I_B}{I_A + I_B}. \quad (9)$$

Parameter	Value
Resonance wavelength / linewidth (FWHM) / depth	1550 nm / 5 nm / 0.8
Sensitivity α	2.5 nm per unit c
Source	Gaussian spectrum, 20 nm FWHM; 150 wavelengths, 1520–1585 nm
DOE	256 $\times$ 256 pixels, 10 μ m pitch; grating period 160 μ m
Input beam radius w_0	200 μ m
Propagation distance z	50 mm
Detectors A, B	300 $\times$ 600 μ m each
Simulation grid	512 $\times$ 512 samples, 5 μ m spacing

Table 1: Simulation parameters.

5 Results

With the grating, the diffracted spot lands at $x = 484.4 \mu\text{m}$, where the grating equation (5) predicts it. Figure 5 shows the broadband detector image with the two detector regions. The spot changes only slightly with concentration: as the dip moves across the source spectrum, the detected light shifts slightly towards shorter wavelengths, and the grating maps this to a small shift of the spot.

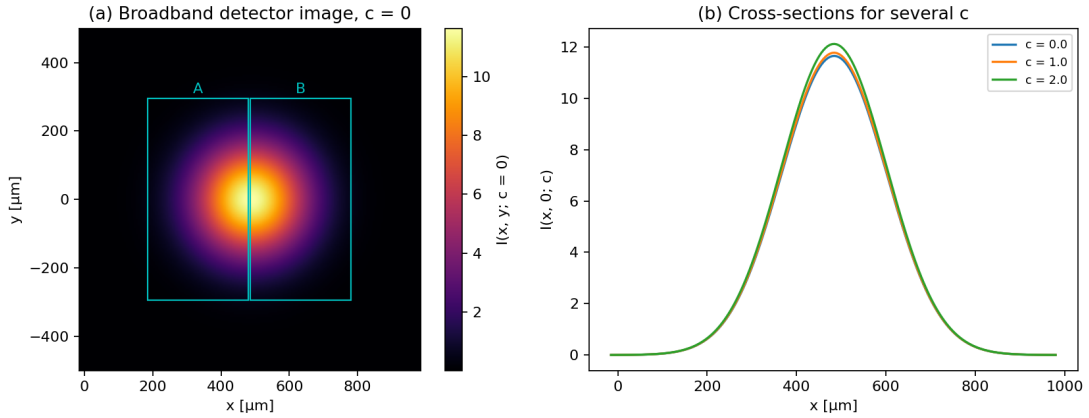


Figure 5: Detector image. (a) Broadband intensity at the detector plane for $c = 0$, with detector regions A and B . (b) Cross-sections for $c = 0, 1$ and 2 .

Figure 6 and Table 2 give the end-to-end result of the forward model. The detector readout and the spot position both respond smoothly and monotonically to concentration. The response of this conventional grating is small: a resonance shift of one linewidth ($c = 1$) changes R by 1.2×10^{-3} and moves the spot by 184 nm. The reason is the grating’s weak dispersion over this footprint: the whole 5 nm linewidth spans only about 1.6 μm at the detector, compared with a spot radius of about 235 μm . This is the baseline that the inverse-designed DOE must improve on in Stage 2.

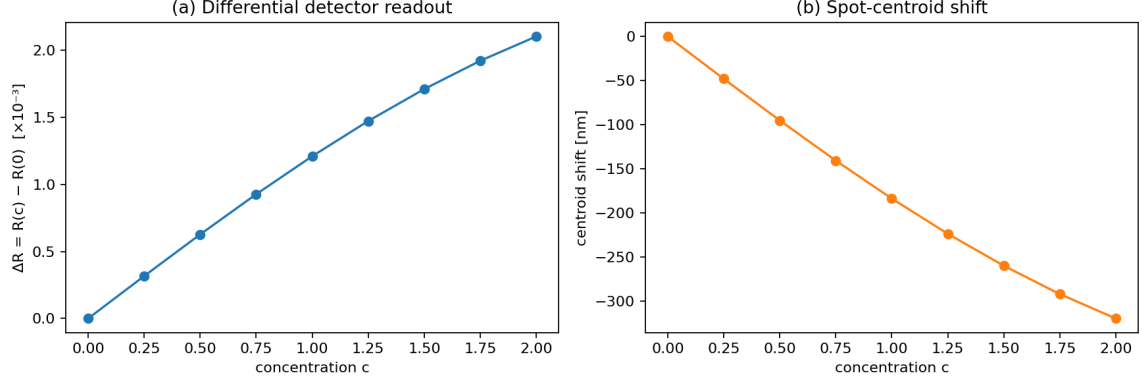


Figure 6: Detector response to concentration with the linear-grating DOE. (a) Change of the differential readout, $\Delta R = R(c) - R(0)$. (b) Shift of the spot centroid.

c	resonance shift [nm]	ΔR	spot shift [nm]
0.0	0.00	0	0
0.5	1.25	0.63×10^{-3}	-95
1.0	2.50	1.21×10^{-3}	-184
1.5	3.75	1.71×10^{-3}	-260
2.0	5.00	2.10×10^{-3}	-320

Table 2: Detector signals from the complete forward model.

6 Validation

Each part of the model was checked against a known result (Table 3). All tests are run by the scripts supplied with the code.

Validation	What is checked	Result
Gaussian propagation	Beam spreading with no DOE vs. the analytic Gaussian beam, 0–150 mm	error 2.4×10^{-6}
Grating equation	Diffracted-spot position vs. Eq. (5), 1450–1650 nm	max. error 1.8 nm (in 480 μ m)
Fraunhofer cross-check	Far-field model vs. angular spectrum at large distance	agree to 0.3% at 2 m
Energy conservation	Optical power through the DOE and propagation	conserved to 10^{-15}
Sampling convergence	Wavelength samples, spatial grid, window and detector pixel size	changes $< 0.2\%$
Completion criterion	Reproducibility, arbitrary spectra and phase masks, gradients for Stage 2	all 7 checks passed

Table 3: Validation of the forward model.

7 Conclusion

A complete forward model has been built that maps analyte concentration to detector signals without a spectrometer. It combines the resonant sensor, a phase-only DOE, angular-spectrum propagation and broadband detection, agrees with analytic results, conserves energy and has converged numerically. With a conventional grating the detector response to a one-linewidth shift is about 10^{-3} , which sets the baseline for the inverse design in Stage 2.

The source code, the step-by-step notebook and instructions to reproduce every figure and number are provided in the accompanying code folder (see its README).